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Optimization of fertilization strategies for paddy fields based on multi-task transfer learning and reinforcement learning

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ABSTRACT

Excessive application of chemical fertilizers triggers severe agricultural non-point pollution while securing crop yields. However, applying a small amount of field trial data to optimize fertilizer management strategies in complex environments is challenging, as yield and environmental risks must be balanced. Here, we develop a hybrid approach combining field trial data, transfer learning, and reinforcement learning (TL-RL) to adjust the fertilizer management algorithm dynamically and autonomously learn the optimal fertilizer application strategy through interaction with the environment. It leverages the effectiveness of field trial data, overcoming the disadvantage that reinforcement learning requires a large amount of data to train the model. We found that the optimal fertilization strategy involved a 13.9 % reduction in nitrogen fertilizer, 10.4 % reduction in phosphorus fertilizer, and 5.44 t/ha of hydrothermal humus charcoal, compared to the traditional fertilization approach. The optimal fertilization scheme resulted in a 7.02 % increase in rice yield, a 20.83 % reduction in total nitrogen concentration, and a 39.13 % reduction in total phosphorus concentration in the paddy surface water. We demonstrate that the TL-RL hybrid model provides a feasible modeling strategy for screening optimal fertilizer application schemes and offers a theoretical approach to achieving sustainable agricultural development by minimizing non-point source pollution while safeguarding crop yields.

1. Introduction

Fertilizer application is essential for increasing global crop yields. However, intensive smallholder farmers often apply excessive fertilizer to ensure crop yields, resulting in significant fertilizer waste and a range of environmental risks (Jote, 2023). Therefore, optimizing fertilizer management practices and seeking the optimal fertilizer application strategy that reduces ecological risks and ensures crop yields has become important in achieving global sustainable development goals. In recent years, methods for reducing fertilizer use with organic materials have attracted considerable attention to control agricultural surface pollution and optimize rice production (Aziz et al., 2018). Among them, hydrothermal humus charcoal (HHC) is a new type of organic material produced from biomass through hydrothermal humification technology, which can improve soil quality and reduce nutrient loss, such as nitrogen and phosphorus. It has excellent potential for application (Zhao et al., 2024). However, determining the optimal ratios of chemical fertilizers and HHC, and optimizing the fertilization strategy to balance rice's nutrient uptake with nitrogen and phosphorus pollution in paddy fields, is an urgent scientific problem to be solved.

Traditional fertilizer application schemes are usually developed through reliable field trials but often require significant manpower, materials, and capital (Dhrubo et al., 2024). Moreover, the trials are conducted in limited contexts, and the collected data cover only specific areas and points in time, making it challenging to reflect the dynamics of the farmland comprehensively (Du et al., 2021). At the same time, when facing multi-objective tasks, manual decision-making is often complex, as it involves weighing and determining the optimal comprehensive benefits. Therefore, efficiently screening fertilization strategies that reflect the dynamic changes in farmland and achieve the optimal

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balance between pollution reduction and yield improvement is an essential challenge for sustainable agriculture (Yin et al., 2021).

Artificial intelligence (AI) models provide ideas for solving the above problems. Reinforcement learning, as an essential learning method in the field of artificial intelligence, has been widely used across robotics, autonomous driving, and agriculture by learning optimal strategies through interaction between intelligence and the environment (Elallid et al., 2022; Devarajan et al., 2023). In agriculture, reinforcement learning has been used to optimize for example irrigation systems and pest control. (Alibabaei et al., 2022; Bashir et al., 2023; Sidiropoulos and Kiourt, 2023; Vibhor et al., 2022). Combining field trials with reinforcement learning can fully leverage the strengths of both and overcome each's shortcomings. On the one hand, field trials can provide high-quality initial data to better construct models based on actual field conditions.

On the other hand, AI can generate more accurate and comprehensive fertilizer recommendations by analyzing and processing field trial data to improve the science and effectiveness of fertilizer decisions (Gul and Banday, 2024; Vinson et al., 2022; Parashar et al., 2024; Khan et al., 2022a,b). However, reinforcement learning typically requires a large amount of data to train the model. When data is insufficient, the search space for intelligence is limited, which may cause the learning process to fall into a local optimum and make it difficult to fully explore the various combinations of states and actions in the environment (Van et al., 2025).

Transfer learning can reduce the training data required for reinforcement learning in new tasks by leveraging the knowledge accumulated by pre-trained models on related tasks (Taylor and Stone, 2009). By combining transfer learning with reinforcement learning, the fine-tuned pre-trained model can be used as part of the reinforcement learning intelligent agent, reducing the cost of reinforcement learning training, enhancing generalization, and accelerating the optimization of reinforcement learning strategies. However, research on applying transfer learning and reinforcement learning to optimize fertilization strategies has not yet been explored. In addition, machine learning models, such as transfer and reinforcement learning, are often regarded as 'black boxes,' and their decision-making processes lack transparency and interpretability (Syed, 2025). As methods for interpreting model predictions, SHAP (Shapley Additive Explanations) and feature importance analysis can reveal insights into each model and its predictions. Predictions can reveal the extent to which each feature contributes to the model's predictions, thereby improving the model's interpretability (Angelopoulos et al., 2023). Therefore, SHAP analysis will be used in this study to identify further key factors affecting environmental risk and yield.

Therefore, the objective of this study is to explore an optimization strategy for HHC to reduce fertilizer use and rationing in paddy fields using a data-driven model that combines multi-task transfer learning and reinforcement learning. Firstly, we applied the source domain data, LSTM (Long Short-Term Memory Network) model and RF (random forest) models to construct the time series and static feature pre-training models, and fine-tuned them with the target domain data; Secondly, the fine-tuned pre-training models were used as part of the reinforcement learning intelligent agent, and the intelligent agent was trained with the target domain data and PPO algorithm to obtain the optimal fertilizer reduction and organic material application plan; Finally, SHAP analysis was used to identify further the key factors affecting environmental risk and yield, which provided a basis for decision-making on precision fertilizer application.

2. Materials and methods

2.1. Data collection and pre-processing

2.1.1. Field data collection

The data for this study were collected from the rice experimental field at Northeast Agricultural University, Heilongjiang Province, China, where a trial was conducted from May to October 2024 to reduce chemical fertilizer use by applying hydrothermal humus charcoal (HHC). (Reference Yang et al., 2019, for the preparation process of HHC.) The rice variety used for the test was Dongnong 430, and the irrigation mode was alternate wetting and drying (see Additional Material for details on the experimental area and irrigation methods). The experiment was set up with three fertilizer application modes: 1) conventional fertilizer application (the fertilizer level was the same as that commonly used in local production: 180 kg/ha N, 90 kg/ha P_2O_5, 90 kg/ha K_2O); 2) fertilizer reduction of 10 % (162 kg/ha N, 81 kg/ha P_2O_5, 90 kg/ha K_2O); and 3) chemical fertilizer reduction of 20 % (144 kg/ha N, 72 kg/ha P_2O_5, 90 kg/ha K_2O). Three HHC application rates (3 t/ha, 4.5 t/ha, 6 t/ha) were established, along with a conventional fertilizer treatment and a control treatment without hydrothermal humus charcoal application. The study employed a completely randomized block-group design with 10 treatments.

During the rice reproductive period, field water was collected in sections along the long edges of the plots. Total nitrogen (TN) and total phosphorus (TP) in the field water were measured using a fully automated intermittent chemical analyzer (model: smartchem2000), with samples being collected 14 times on days 1, 2, 4, 6, 13, 16, 20, 23, 26, 41, 54, 68, 83, and 95 after sowing. Rice yield was measured at the maturity stage. In addition, temperature and precipitation data were collected from the paddy fields during the experimental period, as shown in Fig. 1.

2.1.2. Data pre-processing

The data pre-processing steps include data standardization and normalization. Box plots detect outliers, and all input features are normalized using MinMaxScaler to the [0,1] interval, ensuring consistent magnitudes across features and improving the stability and efficiency of model training. In addition, the data set is split into a training set (80 %) and a test set (20 %), with the training set used for model training and the test set for final model evaluation.

2.2. Model construction and algorithm selection

In this study, we combined multitask transfer learning and reinforcement learning to optimize fertilizer application strategies in paddy fields, reducing surface-source pollution and ensuring rice yield. We use the LSTM (Long Short Term Memory Network) model to learn the time series data features of the source domain, and use RF (Random Forest) to learn the static feature data of the source domain.

The source domain dataset is from the fertilizer reduction trial in Jinhua City, Zhejiang Province, conducted from 2018 to 2019. The

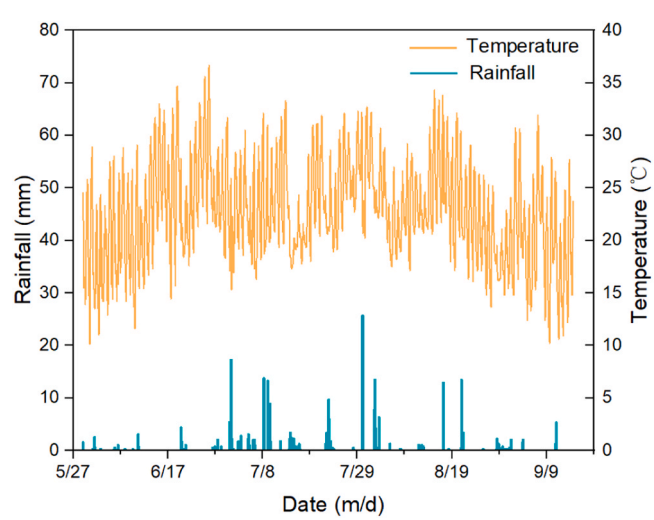

Fig. 1. Temperature and rainfall during the experiment.

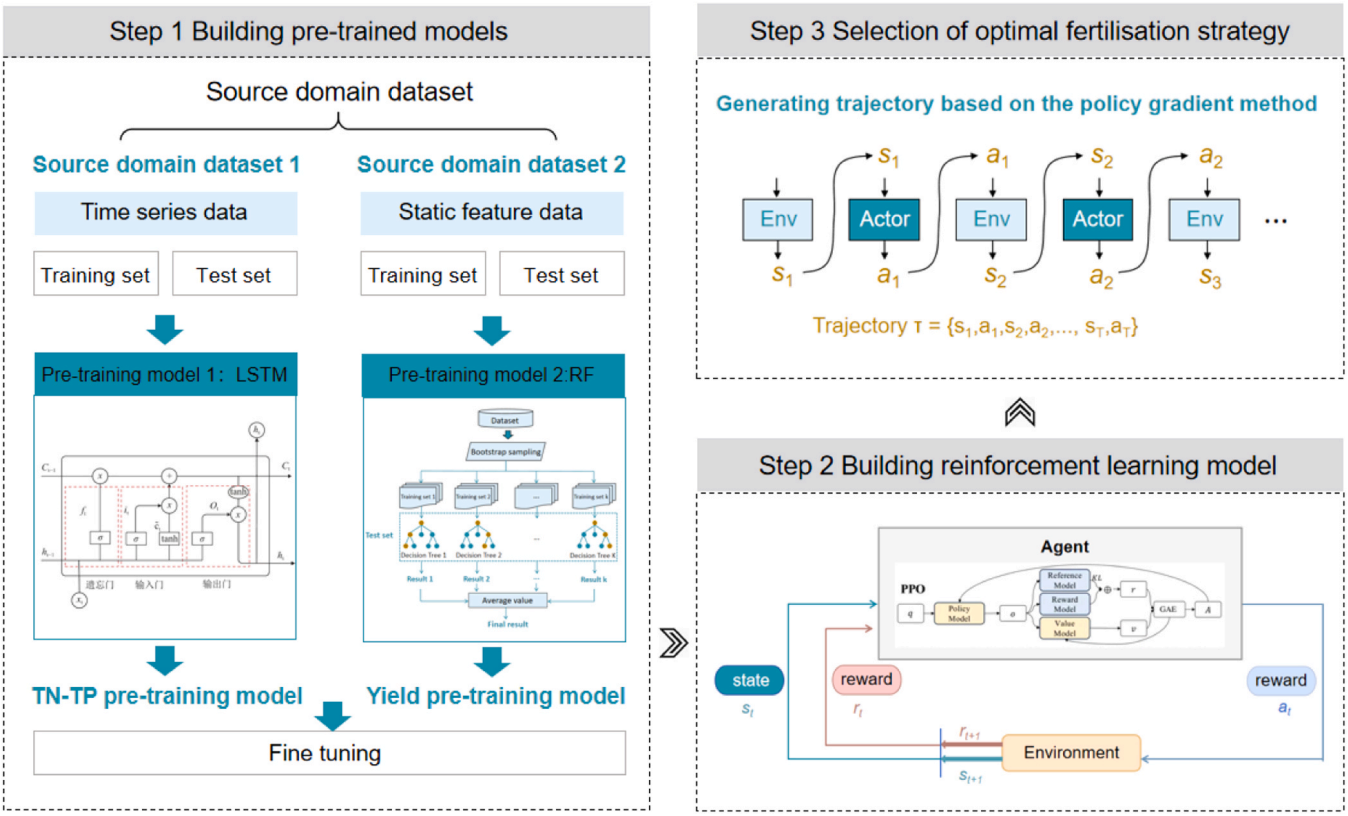

Fig. 2. TL-RL hybrid model construction.

experiment systematically set up no nitrogen fertilizer application, conventional fertilization, gradient nitrogen reduction, and various optimized fertilization treatments (such as adjusting the nitrogen fertilizer base to topdressing ratio, applying slow-release fertilizer, and returning straw to the field), and fully recorded multidimensional data including fertilizer and organic material input, rice yield, plant nitrogen content, and soil background value, to explore ways to reduce nitrogen, increase yield, and increase efficiency.

It is feasible to use Jinhua data as the source domain in this study because there is distribution consistency in the structure of fertilization strategies (both centered on nitrogen reduction), target variables (TN, TP, yield), and pollution yield trade-off relationships, which can provide an effective initial strategy basis for transfer learning. In response to the inevitable domain shift problem between the source domain and the target domain, this model introduces a dynamic reward mechanism (adjusting the weights of pollution and yield according to the training progress), uses adversarial training ideas for feature distribution alignment, expands the diversity of target domain samples using data augmentation techniques, and integrates pre-trained LSTM and random forest models for multi task prediction in environmental design. This enhances the target domain's adaptability to local features while maintaining knowledge transfer from the source domain, effectively alleviating performance degradation caused by unsuitable migration factors, such as geography and climate.

Use an LSTM model to learn time series features in the source domain to construct TP and TN pre-training models. The model's loss function is shown in Eq. (1). In this study, the LSTM model comprises a two-layer network with an output layer. The input layer receives normalized feature data. The LSTM layer has 64 hidden units. A dropout regularization layer is added after each LSTM layer to prevent overfitting, with a dropout rate of 0.2. Finally, the prediction results are output through a fully connected layer (Dense). Subsequently, the pre-trained LSTM model was loaded and fine-tuned using field-measurement data from this experiment for the target task.

$$L_\omega = \frac{1}{N}\sum_{i=1}^{N}\omega_i(y_i' - y_i)^2 \tag{1}$$

Where $L_\omega$denotes the loss value, N is the number of samples in the dataset, $\omega_i$is the weight of each sample, $y_i'$is the predicted value of the model on the i sample, and y_i is the actual value of the model on the first sample.

At the same time, we used the RF model to learn static features from the source domain to construct the yield pre-training model. A two-stage training strategy was employed to build the random forest model. In the first stage, the model was trained on normalized source domain data. In the second stage, the model was fine-tuned using field-measured data from the target domain. The model was constructed using the `RandomForestRegressor` method with 100 trees and a random number generator seeded with 42. Five-fold cross-validation was applied to the target-domain data, with each fold reinitializing the model and repeating the training process described above.

2.3. Interpretability of the model

The SHAP method (Shapley Additive Explanations) explains the decision-making process in complex machine learning models (e.g., LSTM, Random Forest, etc.) by assigning a contribution value to each feature. Specifically, the SHAP value measures the marginal contribution of each feature to the model output for a given prediction, based on all possible feature combinations. For each instance, the SHAP value indicates the feature's contribution to the final prediction, with larger values indicating greater influence. In this study, the SHAP method was used to enhance the model's interpretability by interpreting its predictions and assessing the impact of different features on TN and TP concentrations in the paddy field surface water. At the same time, using the built-in feature importance of the tree model, the feature importance

values for each input variable are calculated by statistically analyzing the contribution of each feature at the splitting node (e.g., impurity reduction or feature usage frequency).

At the same time, to evaluate the robustness of RL algorithms in fertilizer optimization tasks, this study adopts a systematic sensitivity analysis, focusing on the impact of algorithm hyperparameter configurations and reward function weight design on strategy performance. Using the grid search method, the system evaluates the sensitivity of key hyperparameters, including learning rate, training steps, batch size, discount factor, and clipping range. By fixing other parameters and adjusting the target parameters one by one, quantify the impact of each hyperparameter on the strategy's convergence speed and final performance—design weight perturbation experiments to address the core issues of pollution control and yield balance in multi-objective optimization. By adjusting the combination of pollution and yield weights in the system, analyze the sensitivity of the reward function design to the algorithm in multi-objective trade-offs, and ensure that the strategy maintains stable performance across different weight configurations.

2.4. Reinforcement learning and fertilizer application strategy optimization

In this study, we constructed a reinforcement learning-based fertilizer strategy optimization model using a custom `fertilizer Env` environment and a PPO algorithm, where the action space of the environment is the reduction ratio of nitrogen and phosphorus fertilizer (in the range of [10 %, 50 %]) as well as the application amount of organic matters, and the observation space contains meteorological and fertilizer data; and the PPO model adopts the default `MlpPolicy` policy network, set the learning rate to e^{-4}, the number of steps per sample to 2048, the batch size to 128, the number of training rounds to 20, the discount factor to 0.995, the GAE parameter to 0.98, the clip_range to 0.1, the entropy coefficient to 0.001, and the upper limit of gradient paradigm to 0.5, and use ` DummyVecEnv` to encapsulate the environment, set the evaluation callback function, train for 6000 time steps, and finally analyse the model performance by evaluating and visualizing the optimal fertilization strategy and average effect. The PPO algorithm is trained with 20000 time steps, and a dynamic reward function is used to balance paddy pollution control and crop yield targets. Data augmentation techniques are employed to enhance the diversity of training samples. In the evaluation stage, the trained model is run on the same data distribution for 10 rounds of performance validation. The system records multidimensional indicators, including nitrogen and phosphorus reduction ratio, organic fertilizer dosage, total nitrogen and total phosphorus concentration, rice yield, and comprehensive reward value.

Specifically, we simulate the fertilization process by defining an environment (FieldEnvironment class) that receives the current state and fertilization actions and rewards based on changes to the state and actions. After each step, the environment returns to the next state based on the current state and calculates the reward until the maximum number of steps is reached. The model's objective is to guarantee normal rice growth while reducing surface pollution in paddy fields. The optimization problem in reinforcement learning can be formalized as a constrained Markov decision process with a continuous action space :

(1) Decision variables (action space)

The agent outputs a three-dimensional continuous action vector at each decision step:

$$\mathbf{A} = [a_1, a_2, a_3] \tag{2}$$

Where , a_1: N reduction ratio (proportional value, $\in$ [0.1, 0.5]), a_2: P reduction ratio (proportional value, $\in$ [0.1, 0.5]), a_3: Organic fertilizer application rate (kg/ha, $\in$ [0, 6]).

(2) Objective function (reward function)

$$R = \omega_p \cdot (1 - 0.5 \cdot \text{penalty}_{TN} - 0.5 \cdot \text{penalty}_{TP}) + \omega_y \cdot \text{reward}_{Yield} \tag{3}$$

Where , pollution penalty items: $\text{penalty}_{TN} = \max(0, TN - 6.0)^{1.5}$, $\text{penalty}_{TP} = \max(0, TN - 0.6)^{1.5}$. This design imposes a non-linear, increasing penalty on pollution concentrations exceeding the thresholds (*TN*: 6.0 mg/L, *TP*: 0.6 mg/L); the reward yield for output is a piecewise function. Punish when the yield is below 6000 kg/ha; Linearly increase the reward between 6000 kg/ha and the target yield of 9000 kg/ha; If it exceeds 9000 kg/ha, an excess reward will be provided, as shown in formula 3. Dynamic weights ω_p, ω_y: As the training progresses, the pollution weight ω_p gradually decreases from 0.6 to 0.3, while the yield weight ω_y gradually increases from 0.1 to 0.4, guiding the agent to prioritize learning to meet environmental constraints in the early stages of training and then focus on improving yield in the later stages.

$$\text{reward}_{\text{Yield}} \begin{cases} -2 \bullet \dfrac{Y_{\min} - Y}{1000}, & Y < Y_{\min} \\ \dfrac{Y - Y_{\min}}{Y_{target} - Y_{\min}}, & Y_{\min} \le Y < Y_{target} \\ 1 + 0.5 \bullet \dfrac{Y - Y_{target}}{1000}, & Y \ge Y_{target} \end{cases} \tag{4}$$

(3) Constraints:

Environmental constraints: total nitrogen concentration $TN \le 6.0$ mg/L, total phosphorus concentration $TP \le 1.0$ mg/L, rice Yield ≥ 5000 kg/ha. Any behavior that violates these constraints will impose significant penalties (-5 or −3) in the reward function.

Action space constraint: The range of decision variables is directly guaranteed by the action space of the agent, as described above.

(4) State variables (State space)

State space: The environment's observation state is a 6-dimensional vector comprising three meteorological factors (temperature, precipitation, humidity) and three fertilization-related factors (nitrogen fertilizer application, phosphorus fertilizer application, organic fertilizer application).

(5) System Dynamics: A pre-trained LSTM model is used to predict TN and TP concentrations under fertilization strategies, and a random forest model is used to predict rice yield, serving as an alternative model for simulating environmental dynamics.

Note: The parameters in this study are divided into three categories: the first category is the core hyperparameters of the optimization algorithm (such as learning rate, discount factor, network structure), which are fixed by the algorithm and determined through pre research, and users do not need to adjust them; The second type is user-defined policy parameters, including environmental thresholds (such as TN/TP target concentration, target yield) and action space boundaries (such as upper and lower limits of fertilizer application), which allow users to customize according to specific production goals and environmental regulations; The third type is randomness parameters, mainly including random processes in network initialization and data augmentation, which ensure the reproducibility of experimental results by setting random seeds.

(6) Adam optimizer

To optimize the loss function and improve model training efficiency, we use the Adam optimizer, a gradient-based optimization algorithm that combines the Momentum and Adaptive Learning Rate approaches, offering high computational efficiency and low memory requirements. The updated Equation for the Adam optimizer 5–9:

$$\boldsymbol{m_t} = \beta_1 \boldsymbol{m_{t-1}} + (1-\beta_1)\nabla_\theta L(\boldsymbol{\theta}) \tag{5}$$

$$\boldsymbol{v_t} = \beta_2 \boldsymbol{v_{t-1}} + (1-\beta_2)\nabla_\theta L(\boldsymbol{\theta})^2 \tag{6}$$

$$\boldsymbol{m_t}' = \frac{\boldsymbol{m_t}}{1-\beta_1{}^t} \tag{7}$$

$$\boldsymbol{v_t}' = \frac{\boldsymbol{v_t}}{1-\beta_2{}^t} \tag{8}$$

$$\boldsymbol{\theta_{t+1}} = \boldsymbol{\theta_t} - \frac{\alpha \boldsymbol{m_t}'}{\sqrt{\boldsymbol{v_t}' + \epsilon}} \tag{9}$$

Where m_t is the first-order moment estimate of the gradient (momentum term), v_t is the second-order moment estimate of the gradient (variance term), $m_{t-1}$and v_{t-1} denote the first-order moment and second-order moment estimates of the gradient at the previous time step (i.e., step $t-1$), respectively, β_1 and β_2 are the decay rate of the momentum and the decay rate of the squared gradient. $\nabla_\theta L(\theta)$ denotes the gradient of

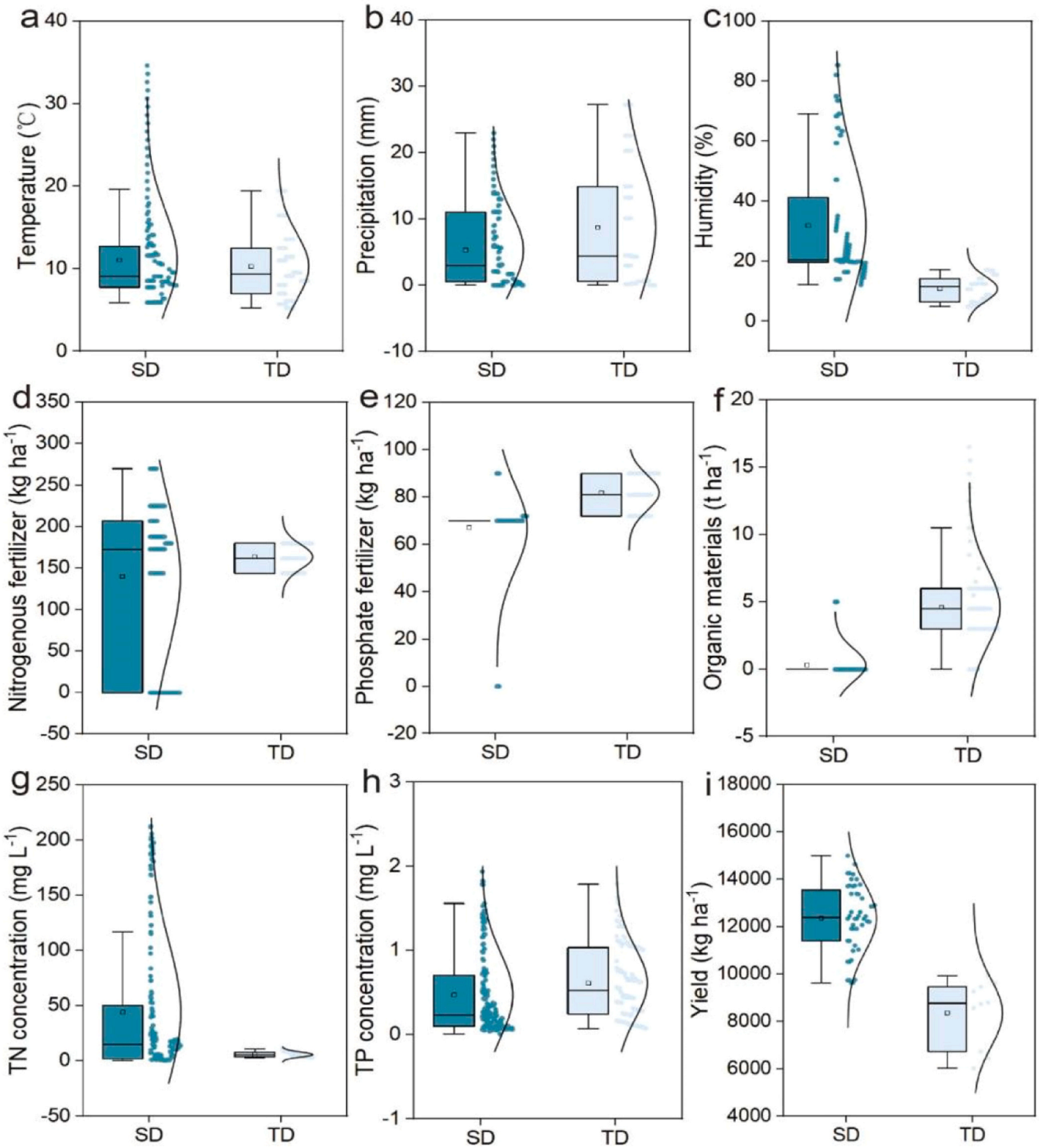

Fig. 3. Distribution of datasets in source and target domains. Note: SD represents the source domain, TD represents the target domain.

the loss function $L(\theta)$ with respect to the parameter θ, m_t' and v_t' are the first-order moment estimate and second-order moment estimate of the gradient corrected for bias, respectively, and θ_{t+1} is the updated value of the model parameter at step $t+1$, θ_t is the current value of the model parameter at step t. α is the learning rate, where ϵ is a small constant, usually set to 10^{-8} to prevent the denominator from going to zero and to ensure computational stability.

2.5. Model evaluation criteria

To assess the performance of the model, the following evaluation metrics were used in this study:

(1) Mean Squared Error (MSE)

$$\mathrm{MSE} = \frac{1}{\mathrm{N}} \sum_{i=1}^{\mathrm{N}} (y_i - y_i')^2 \tag{10}$$

Where N is the number of samples in the dataset, y_i is the actual value of the model on the ith sample, and y_i' is the model's predicted value on the sample.

(2) Mean Absolute Error (MAE)

$$\mathrm{MAE} = \frac{1}{\mathrm{N}} \sum_{i=1}^{\mathrm{N}} |y_i - y_i'| \tag{11}$$

(3) R-squared (R^2)

$$\mathrm{R}^2 = 1 - \frac{\sum_{i=1}^{\mathrm{N}} (y_i - y_i')^2}{\sum_{i=1}^{\mathrm{N}} (y_i - y_i'')^2} \tag{12}$$

Where y_i'' is the average of the actual values.

3. Results and discussion

3.1. Data preparation and model prediction performance

3.1.1. Comparison of source and target domain data

In this study, the feasibility of transfer learning in ecological modeling was systematically assessed by Fig. 3. The analysis showed that the two domains had significant distribution overlap (IQR overlap > 65 %) in parameters such as temperature and rainfall, which provided a reasonable basis for feature migration, while parameters such as humidity, N fertilizer, organic matters, and yield showed significant differences. Therefore, this study adopts a hierarchical migration strategy based on the dataset's distributional characteristics: parameters with high overlap are directly migrated, those with moderate differences are transformed using a linear transformation algorithm, and those with significant differences are combined using a subdomain-adaptive algorithm.

3.1.2. Performance of predicting TP, TN, and yield after fine-tuning

Fig. 4a-b presents a comparison between the actual values and predicted values of TN concentration, TP concentration, and rice yield. The close alignment between the predicted and exact values demonstrates the robustness of the model in estimating these critical agricultural parameters. The fitness of predicted TP, TN, and yield to the actual values by applying the transfer learning approach on the target domain is presented in Fig. 4d-e. The fine-tuned pre-trained model showed high explanatory power in predicting TP, TN, and yield, with R^2 values of 0.8, 0.75, and 0.89, respectively, indicating a strong correlation between the expected and actual values. The MAEs of 0.67, 0.14, and 324.15 and the RMSEs of 0.87, 0.17, and 455.26, respectively, further confirmed the transfer learning model's prediction accuracy.

3.2. Reinforcement learning training process

Fig. 5a illustrates the variation in the reward curve with the number

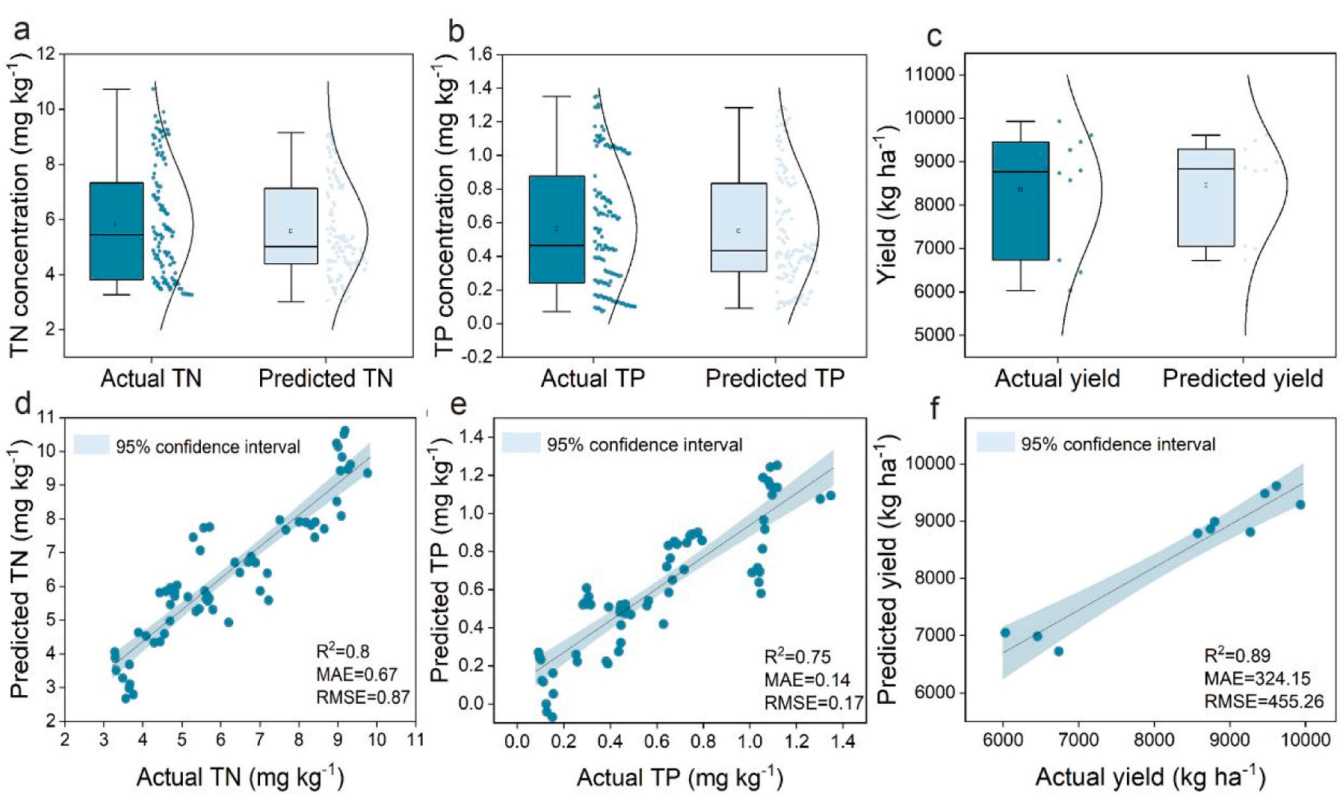

Fig. 4. Distribution and fitting of tested and actual values.

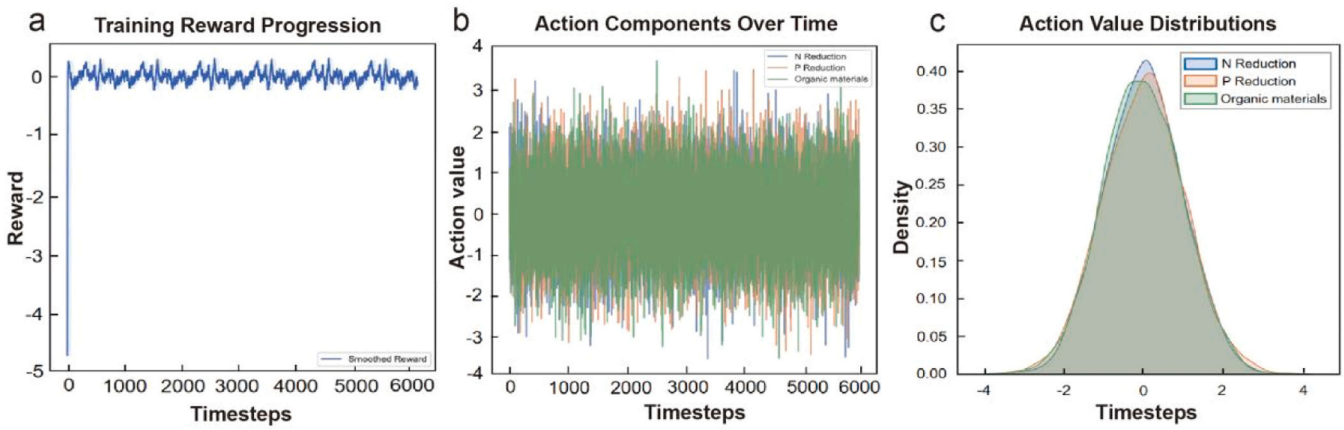

Fig. 5. Reinforcement learning training process. a is the training reward curve; b is the dynamic optimization feature of the fertilization strategy; c is the action value distribution feature.

of training steps. The large fluctuations at the beginning of the curve (0–1000 time steps) indicate that the intelligent agent has undergone a trial-and-error process in exploring different fertilization strategies. The decrease in reward value may stem from two extreme cases: i. excessive reduction of fertilizer application leads to a significant decline in yield, pulling down the reward; ii. The improper rationing of organic fertilizer fails to reduce surface source pollution effectively, and the environmental goal is not met. This stage aligns with the expected behavior of reinforcement learning: accumulating experience through extensive exploration and laying the foundation for subsequent strategy optimization (Saglam and Kozat, 2023). As training progresses (1000–6000 time steps), the reward curve gradually smooths and stabilizes, indicating that the intelligent body successfully learns a fertilizer strategy that balances yield and pollution. This result verifies the effectiveness of the PPO algorithm: it avoids drastic updates by clipped objectives and achieves stable strategy improvement. The final strategy involves a moderate reduction of chemical fertilizer combined with the substitution of organic materials, aligning with the conclusion of 'organic-inorganic synergy' in agricultural research (Li et al., 2025; Shang et al., 2025). In Fig. 5b, the time series of the three decision variables —nitrogen fertilizer reduction, phosphorus fertilizer reduction, and organic material application —are shown, revealing the optimal fertilizer application strategy gradually formed by the intelligent body during training. Each decision variable exhibits steep fluctuations, indicating that the intelligent body explores various combination schemes during the initial stage. During the strategy's convergence stage, the three curves exhibit regular oscillations, indicating that the strategy tends to stabilize and that the reward value increases gradually (Fig. 5b).

Fig. 5c shows the probability density distributions of the three decision variables—nitrogen fertilizer reduction (N Reduction), phosphorus fertilizer reduction (P Reduction), and organic materials application —revealing key patterns in strategy optimization for the intelligentsia. Nitrogen fertilizer reduction showed a single peak and a tall distribution (peak density 0.38 ± 0.02), with 90 % of the decisions concentrated in the [-1.8,1.8] interval and the tail decaying rapidly ($|x|>2$ probability <3 %). This indicates that, in simulated agricultural systems, the agent learns that nitrogen fertilizer application rate must be precisely controlled within a relatively narrow "optimal window". Deviation from this window (whether too much or too little) will lead to a significant decrease in the overall reward (yield or environmental penalty); therefore, the agent's strategy distribution exhibits low variance and high concentration. This reflects the high sensitivity and strong constraint of nitrogen management in the system. Phosphorus fertilizer reduction showed a broad-peaked distribution, with a wider decision range, and the probabilities of the three decision variables were higher. The phosphorus fertilizer reduction showed a broad peak distribution with a wider decision range (90 % in [-3.2,3.2]). This indicates that the constraints on phosphorus fertilizer application learned by the agent are relatively loose, allowing for adjustments on a larger scale without having a decisive negative impact on the ultimate goal. This may suggest that the "buffering" characteristics of phosphorus dynamics in simulated environments or their marginal impact on yield vary smoothly at different levels (Cui et al., 2024 ; Bhatia et al., 2025); the application of organic matter showed a unique three-peak structure (with the peaks located at $x \approx -0.2$, 0, and +0.6). This can be explained as the agent learning the discontinuous effects or threshold characteristics of organic fertilizer application. These distributional features jointly verified the study's ability to optimize multiple objectives (balancing yield and environmental goals through distributional skewness) and to identify spatial and temporal heterogeneity in nutrient management. Translating these observations into practical application, we can deduce that nitrogen fertilizer needs to be precisely regulated, while phosphorus fertilizer can be moderately relaxed. The threshold strategy is suitable for organic management and provides a theoretical basis for parameter setting in intelligent fertilizer application systems.

3.3. Numerical results of the optimal strategy

In this study, three-dimensional response surface plots demonstrated the combined effects of fertilizer reduction and organic matter application on TN and TP concentrations and rice yield in paddy fields (Fig. 6). TN and TP concentrations decreased with increasing N and P fertilizer reduction ratios. However, rice yield gradually reduced once the fertilizer reduction ratio exceeded a threshold. The optimal solution, obtained using the PPO algorithm, reduced surface pollution in the paddy field while maintaining rice yield, with 13.9 % reductions in N fertilizer and P fertilizer, and 5.44 t/ha in organic matter (HHC). The differences in N and P reductions reflect the specificity of rice's nutrient requirements: the higher N reductions could be attributed to the high N background of local soils, and the conservative P reductions are in line with P fixation by soil (Wang et al., 2024 ; Boostani et al., 2025). The application of 5.44 t/ha of organic matter reflects the 'partial substitution' strategy, which balances the regulation of carbon and nitrogen ratios with the rapidity of fertilizer supply.

Under this strategy, surface pollution was significantly reduced. Compared with the conventional fertilizer application method, the TN concentration decreased from 6.53 mg/L to 5.17 mg/L (a 20.83 % decrease); the TP concentration decreased from 0.69 mg/L to 0.42 mg/L (a 39.13 % decrease). Moreover, by applying the optimized fertilizer application scheme, TN and TP concentrations were reduced more quickly to below the nitrogen and phosphorus threshold for farmland withdrawal water (5 mg/L for total nitrogen and 0.3 mg/L for total

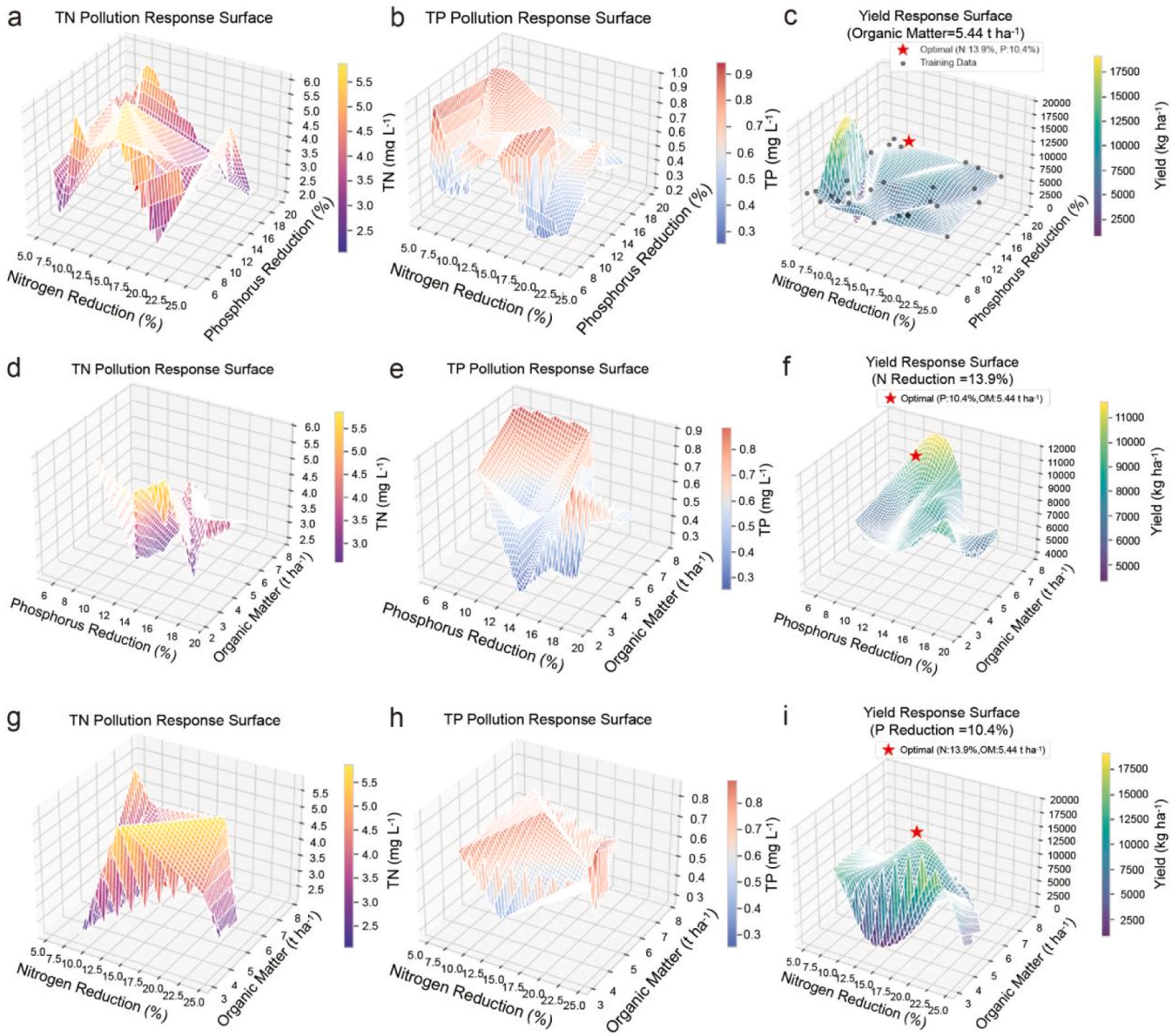

Fig. 6. Three-dimensional response surface plots of TN, TP, and yield. Note: Fig. 6a-c show the three-dimensional response surfaces of TN, TP, and yield with nitrogen reduction and phosphorus reduction when organic matter application is fixed at the optimal level (organic matter = 5.44 t/ha). Fig. 6d-f show the three-dimensional response surfaces of TN, TP, and yield with phosphorus reduction and changing organic matter when nitrogen fertilizer application is fixed at the optimal level (N reduction = 13.9 %). Fig. 6g-i show the three-dimensional response surfaces of TN, TP, and yield with nitrogen reduction and organic matter when phosphorus fertilizer application is fixed at the optimal level (P reduction =10.4 %). In addition, the organic matter in this figure is HHC.

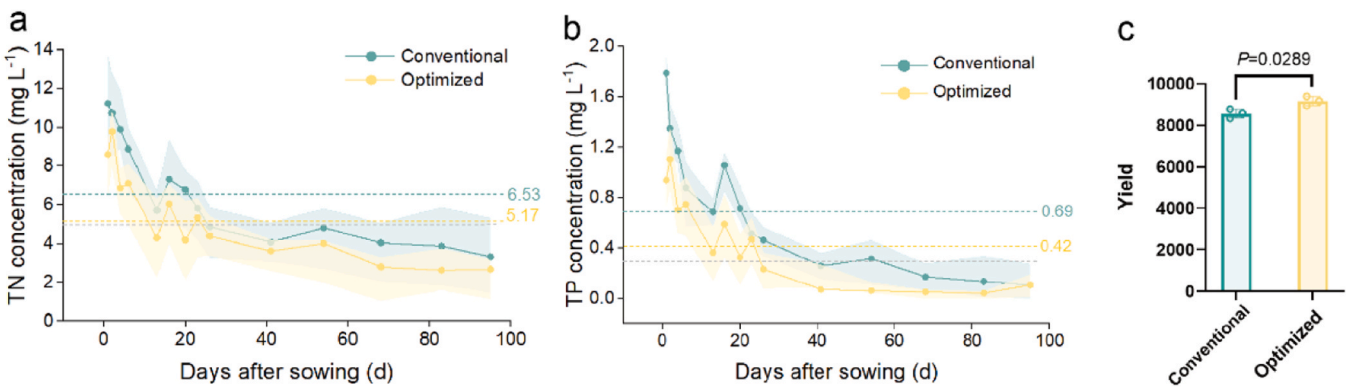

Fig. 7. TN, TP, and yield between conventional and optimized fertilization schemes.

phosphorus) (Standard for irrigation water quality GB, 5084–2021, 2021). In addition, the yield was increased by 7.02 % compared with the traditional fertilizer application method, achieving synergistic optimization of yield maintenance and surface pollution control (Fig. 7).

3.4. Model evaluation

The evaluation scheme adopted in this study effectively assesses the strategy's generalization ability and prevents overfitting. Specifically, the trained strategy was evaluated in a completely independently initialized environment instance and underwent 10 rounds of independent testing, with the strategy running in deterministic mode during the evaluation. This method ensures that the evaluation process is not affected by the internal state of the training environment and can objectively reflect the performance of the strategy under the same data distribution but different initial conditions and decision sequences. The current evaluation design has effectively demonstrated the strategy's generalization ability through environment instance isolation, multiple rounds of independent testing, and deterministic strategy execution, thereby avoiding overfitting to the training environment.

3.4.1. Comparison with Bayesian model

In addition, we systematically compared the proposed TL-RL model with advanced Bayesian optimization methods in precision agriculture. The results show that the TL-RL method exhibits unique advantages in dealing with complex multi-objective trade-off problems: although the comprehensive reward function value is low, it discovers a more environmentally breakthrough strategy - significantly reducing the total nitrogen (TN) concentration from 6.84 mg/L optimized by Bayesian optimization to 5.81 mg/L, highlighting the optimization ability of TL-RL under complex environmental constraints. This leap in environmental performance was achieved at the cost of minimal yield loss (only 0.4 %)

In terms of agricultural inputs, the strategy proposed by TL-RL is more efficient: nitrogen fertilizer application is reduced by 50 %, far higher than the 10 % reduction achieved by Bayesian optimization. Although the comprehensive reward function value of TL-RL is relatively low, its performance in environmental goals is better, indicating that traditional reward function designs may not fully capture the long-term value of environmental benefits. In addition, the TL-RL strategy showed a slight improvement in phosphorus reduction, although the effect was not significant, further confirming the method's potential for comprehensive pollutant control.

Although the TL-RL strategy resulted in a 0.4 % loss in crop yield, given its dual benefits of a significant reduction in nitrogen fertilizer and a significant improvement in environmental quality, this small cost is acceptable. The "low input- high environmental benefit" strategy revealed by TL-RL provides a new path for achieving sustainable agricultural development under strict environmental regulations. The result chart is shown in Supplementary Material 4.

The findings of this study further support the value of introducing reinforcement learning methods into agricultural system optimization, especially in high-dimensional, nonlinear, and multi-objective decision-making scenarios. Future research can further explore the redesign of reward functions to more accurately balance production and environmental goals, while extending the TL-RL method to more diverse environmental indicators and crop systems.

3.4.2. Calculate cost and efficiency

The algorithm proposed in this article has fully considered its practicality in design and ensured computational efficiency in the following aspects:

Firstly, the core of the algorithm is a forward prediction process. In actual deployment, the trained RL agent only needs to perform a single forward pass to make decisions, with minimal computational overhead. It mainly consists of a small neural network (256 ×2) and two lightweight prediction models (LSTM and RF) for inference. This process can be completed in milliseconds on a regular CPU, fully meeting the real-time decision-making needs in the field.

Secondly, although the offline training phase involves model prediction and strategy iteration, it does require some computing resources, but this is a one-time investment. We adopted PPO, an algorithm that excels in both sample efficiency and stability. We set a moderate number of training steps (20000) so that the total training time stays within a few hours on a single GPU or a high-performance CPU, which is entirely acceptable for planning an agricultural season.

3.4.3. Sensitivity analysis of RL parameters and reward function weights

The effective RL configuration for multi-objective fertilization optimization was determined through a parameter sensitivity analysis, which revealed that the intelligent agent exhibited stable optimization performance in both pollution control and yield balance. The specific results are shown in Supplementary Material 5.

3.5. Interpretability of the model

3.5.1. SHAP analysis

This study explored the key factors affecting TN, TP, and rice yield through characteristic importance analysis and SHAP value analysis. Fig. 8a, c, and e show the characteristic importance analysis, which revealed that precipitation was the most critical factor affecting TN, with a characteristic importance of 0.356, followed by temperature (0.327) and humidity (0.312). For TP, temperature was the most important influencing factor, with a characteristic importance of 0.181, followed by precipitation (0.173) and humidity (0.170). For rice yield, N fertilizer (0.437) and P fertilizer (0.411) were the two most influential factors, with an importance of 0.149 for HHC.

Further SHAP value analyses (Fig. 8b, d, and f) revealed the specific effects of these characteristics on the target variables. Nitrogen and phosphate fertilizer application for TN and TP presented significant positive SHAP values, confirming that they are the main controlling factors for TN and TP loss. Humidity, rainfall, and HHC presented significant negative SHAP values, indicating that these factors inhibit pollutant loss. This may be because high humidity maintains soil pore water continuity and promotes the infiltration of pollutants to the deeper layers (rather than surface runoff) (Han et al., 2025). Meanwhile, humus carboxyl/phenolic hydroxyl groups formed stable complexes with PO_4^{3-}, thereby enhancing the ion-exchange capacity for NH_4^+ and promoting the formation of soil aggregates, thereby reducing the risk of TN and TP loss in runoff (Xu et al., 2021). In conclusion, under specific precipitation and temperature conditions, optimizing fertilizer and HHC application strategies is crucial for regulating TN and TP losses, thereby ensuring rice yields while reducing surface-source pollution in paddy fields and promoting sustainable agricultural development.

3.5.2. Strategic distillation

This study is based on the RL strategy distillation method. It extracts three key decision rules for nitrogen fertilizer reduction, phosphorus fertilizer reduction, and organic fertilizer application from 3000 state-action pairs. Analysis shows that the nitrogen fertilizer reduction strategy is mainly dominated by temperature (54.48 % importance), and the decision-making is relatively unified. When the temperature is below 9.35, a reduction of 0.5 is recommended (i.e. a 50 % reduction); The reduction of phosphorus fertilizer is highly dependent on precipitation (88.94 % importance), and the magnitude of reduction increases significantly with the increase of precipitation, especially under high phosphorus fertilizer usage and high organic matter content conditions, reaching 0.49; The application of organic fertilizer is significantly influenced by both nitrogen fertilizer usage (47.53 %) and temperature (37.72 %), and is only recommended when the temperature is above 15.04 and the nitrogen fertilizer usage is moderate. The analysis of feature importance reveals that humidity has no substantial impact on

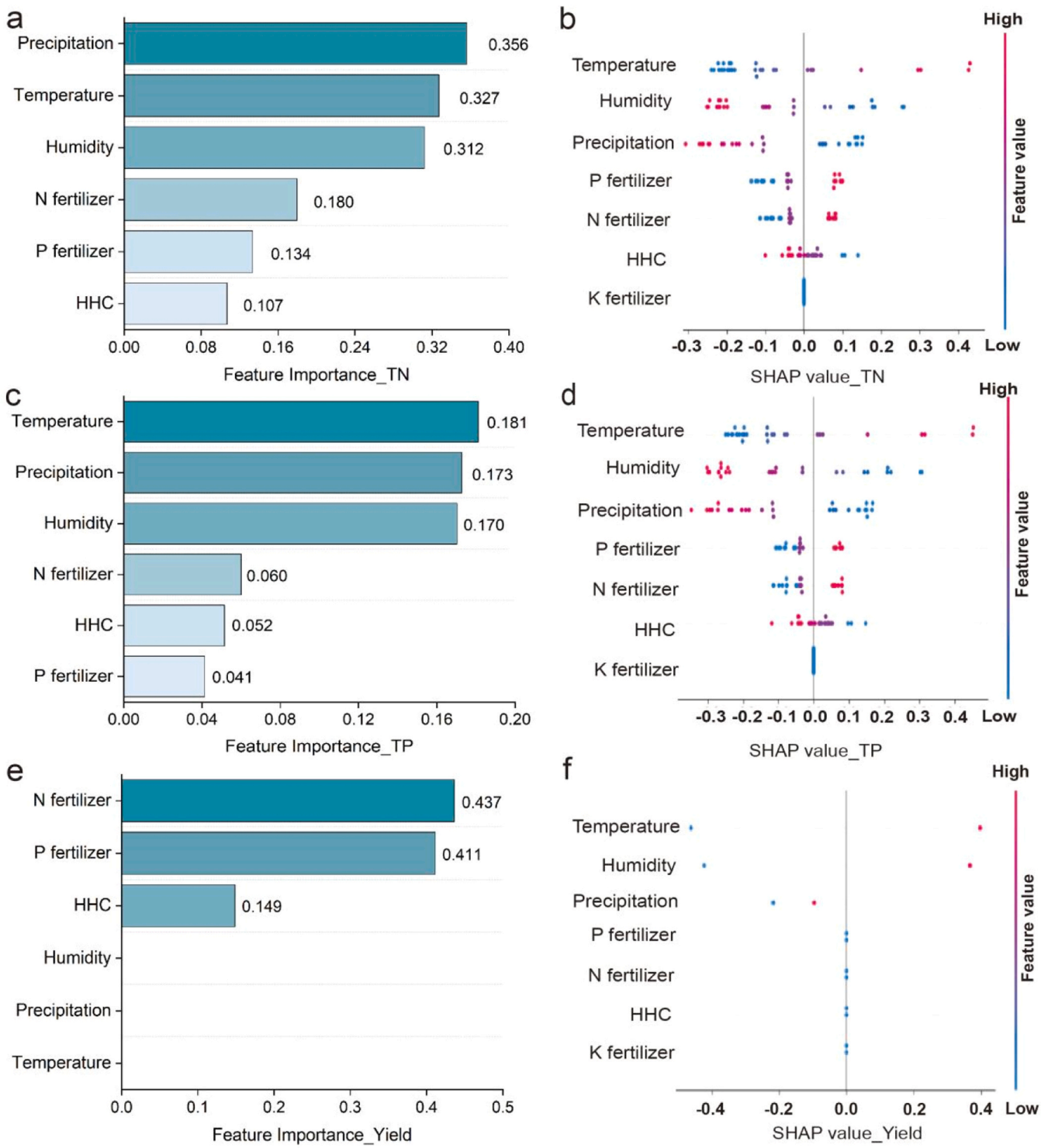

Fig. 8. Importance of Features and SHAP Analysis.

the three decisions, providing clear guidance for sensor deployment and state-space design in agricultural intelligent decision-making systems. That is, nitrogen fertilizer strategies should focus on monitoring temperature and organic matter content, phosphorus fertilizer strategies should concentrate on precipitation changes, and organic fertilizer strategies should consider the synergistic effect of nitrogen fertilizer use and temperature conditions simultaneously (see Supplementary Material 7 for the figure).

Overall, compared with existing decision-making frameworks, this decision-making system explores the long-term effects of different fertilization strategies through dynamic interactions and directly outputs the optimal fertilization plan, addressing the shortcomings of existing methods, which can only analyze but cannot make decisions. At the same time, the dynamic balancing mechanism's reward function addresses the shortcomings of traditional fixed-weight methods. It not only provides specific fertilizer ratio schemes but also visualizes the evolution of strategies through multidimensional visualization, offering transparent, operable decision support for agricultural management and significantly enhancing the system's practical value.

3.6. Limitations and future directions

To deploy the intelligent fertilization strategy of this study in the real world, a feasible process is as follows: Firstly, by deploying field IoT sensors (such as temperature, humidity, and rainfall sensors) and integrating historical records, a data collection system for real-time collection of meteorological, soil, and fertilization data is constructed, and data from different growth seasons or geographical regions is used as a completely independent external test set. Then, the trained PPO model is packaged as a cloud API or deployed to field-edge computing equipment to complete model deployment. In the decision support stage, the system dynamically recommends an optimal fertilization plan, including the N/P reduction ratio and organic fertilizer dosage, based on real-time perceived environmental data. This scheme can directly generate variable fertilization prescription maps, which are automatically executed by existing agricultural machinery that supports variable rates, such as precision fertilizing machines that apply organic fertilizers. Finally, by establishing a feedback loop, actual TN, TP, and yield data are collected after the fertilization season to enable continuous online updating and optimization of the model, thereby forming a closed-loop precision agriculture management system.

However, this deployment path may face many challenges: lagging real-time monitoring technology for key water quality parameters leads to delayed feedback data, and variations in the physical properties of organic fertilizers affect the accuracy of variable fertilization. The lack of fidelity in the simulation environment used by this research institute, along with parameter sensitivity, is the key constraint. The simulator simplifies hydrological, biogeochemical, and crop physiological dynamics in complex agricultural ecosystems, resulting in a significant gap between the training strategy and the real world. It is challenging to fully capture dynamic factors, such as extreme climate-driven feedbacks, soil microbial feedback, and field heterogeneity, which can lead to poor performance even for excellent strategies in real-world simulations. In the future, systematic simulator calibration should be conducted by incorporating physical mechanism models and real-world data. Only by making breakthroughs in sensing technology, agricultural machinery operations, and algorithm safety can we achieve a successful transition from simulation to the field.

4. Conclusions

In this study, we develop a hybrid approach combining field trial data, transfer learning, and RL to adjust the fertilizer management method dynamically and autonomously learn the optimal fertilizer application strategy through interaction with the environment. It fully leverages the effectiveness of field trial data, overcoming the disadvantage that RL requires a large amount of data to train the model. The results showed that, compared with conventional fertilization, the 13.9 % reduction in nitrogen fertilizer and the 10.4 % reduction in phosphorus fertilizer achieved with the dosing of 5.44 t/ha HHC could synergistically optimize yield maintenance and reduce non-point source pollution in paddy fields. Meanwhile, SHAP analysis confirmed that optimizing fertilizer and HHC application strategies was a key factor in regulating TN and TP losses and rice yield under certain precipitation and temperature conditions. In the future, sensor data or long-term field monitoring can be combined to improve the realism of the model inputs, introduce more environmental indicators to construct a more comprehensive reward function, and consider the effects of the differences in soil and climate conditions in different regions on the fertilizer application strategy, further improving the cross-regional generalization ability of transfer learning.

CRediT authorship contribution statement

Fan Yang: Supervision, Funding acquisition. **Giulia Ischia:** Validation. **Ying Zhao:** Writing – review & editing, Supervision. **Kui Cheng:** Validation. **Zhuqing Liu:** Validation. **Zeyu Zhang:** Writing – original draft, Formal analysis, Conceptualization. **Dongxing Xie:** Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.agwat.2025.109965.

Data availability

Data will be made available on request.

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